

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 2: Center of Gravity Theory, Datums & Law of the Lever

Syllabus Source: CAE Oxford (pp. 23-42) | Pass Target: $\geq 90\%$ AviationExam

1. CENTER OF GRAVITY (CG) PHYSICAL CONCEPT

The Center of Gravity (CG) is the single imaginary point through which the resultant force of gravity acts on the entire aircraft. If suspended by a cable attached precisely to its CG, the aircraft would remain in perfect horizontal equilibrium in any attitude.

2. THE DATUM & ARM SIGN CONVENTIONS

To calculate where the CG is located, all physical positions are measured from an arbitrary reference plane called the DATUM:

- The Datum: A vertical reference plane perpendicular to the longitudinal axis, chosen by the aircraft designer. Common locations: (1) Aircraft nose tip, (2) Engine firewall, (3) Wing leading edge.
- The Arm (Lever Arm): The horizontal distance from the datum to the center of gravity of an individual item (passenger seat row, cargo compartment, fuel tank).
- Sign Convention: By universal convention: Distance BEHIND (aft of) the datum is POSITIVE (+). Distance IN FRONT of (forward of) the datum is NEGATIVE (-).
- The Moment Equation: Moment = Mass x Arm. Expressed in kg.m, kg.mm, or lb.in.

3. THE LAW OF THE LEVER & CENTER OF GRAVITY FORMULA

The total moment of an aircraft about the datum is equal to the algebraic sum of the individual moments of all items loaded onto it:

- Master CG Equation: Aircraft CG Location = (Total Moment) / (Total Mass) = $\text{Sum}(\text{Mass}_i \times \text{Arm}_i) / \text{Sum}(\text{Mass}_i)$.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CG Calculation via Moments Table

Problem: Basic Empty Mass = 30,000 kg at Arm 10 m. Pilot = 80 kg at Arm 4 m. Cargo = 2,000 kg at Arm 6 m. Where is the new CG? Step 1: Build table: Item | Mass | Arm | Moment (Mass x Arm): - Aircraft: 30,000 kg x 10 m = 300,000 kg.m - Pilot: 80 kg x 4 m = 320 kg.m - Cargo: 2,000 kg x 6 m = 12,000 kg.m Step 2: Sum Total Mass = 30,000 + 80 + 2,000 = 32,080 kg. Step 3: Sum Total Moment = 300,000 + 320 + 12,000 = 312,320 kg.m. Step 4: Calculate CG = Total Moment / Total Mass = 312,320 / 32,080 = 9.736 meters aft of datum.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Aircraft Center of Gravity from Loading Table

SCENARIO (Classic EASA Exam Question): An aircraft has the following loading schedule with the Datum located at the Engine Firewall: - Basic Empty Mass: 1,200 kg with an Arm of +1.50 m - Pilot and Copilot: 160 kg with an Arm of +1.10 m - Rear Passengers: 150 kg with an Arm of +2.20 m - Baggage in Nose Compartment: 40 kg with an Arm of -0.80 m (AHEAD of firewall!) - Fuel (in wings): 180 kg with an Arm of +1.60 m

QUESTION: Calculate the exact position of the aircraft's Center of Gravity relative to the Datum. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Build the Mass x Arm = Moment Table: - Item 1 (BEM): $1,200 \text{ kg} \times (+1.50 \text{ m}) = +1,800.0 \text{ kg.m}$ - Item 2 (Front): $160 \text{ kg} \times (+1.10 \text{ m}) = +176.0 \text{ kg.m}$ - Item 3 (Rear): $150 \text{ kg} \times (+2.20 \text{ m}) = +330.0 \text{ kg.m}$ - Item 4 (Nose Bag): $40 \text{ kg} \times (-0.80 \text{ m}) = -32.0 \text{ kg.m}$ [NOTE: NEGATIVE MOMENT!] - Item 5 (Fuel): $180 \text{ kg} \times (+1.60 \text{ m}) = +288.0 \text{ kg.m}$

Step 2: Sum the Total Mass: - Total Mass = $1,200 + 160 + 150 + 40 + 180 = 1,730 \text{ kg}$. Step 3: Sum the Algebraic Moments (be careful with the negative sign!): - Total Moment = $1,800.0 + 176.0 + 330.0 - 32.0 + 288.0 = +2,562.0 \text{ kg.m}$. Step 4: Divide Total Moment by Total Mass: - CG = Total Moment / Total Mass = $+2,562.0 \text{ kg.m} / 1,730 \text{ kg} = +1.4809 \text{ m}$. FINAL ANSWER: The Center of Gravity is located at +1.48 m (1.48 meters aft of the firewall).

[!] EXAM TRAP (AVIATIONEXAM) - The Forward Datum / Negative Arm Trap

- If an item is placed IN FRONT of the datum, its arm is NEGATIVE (-). - Its moment is therefore NEGATIVE! (Mass x (-Arm) = -Moment). - In the total moment sum, you MUST SUBTRACT this negative moment, but you MUST STILL ADD its mass to the total mass!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the definition of the 'ARM' of a load in mass and balance calculations?**

- [A]: The vertical height of the item above the cabin floor
- [B]: The horizontal distance from the reference DATUM to the center of gravity of the item
- [C]: The length of the fuselage
- [D]: The structural thickness of the wing spar

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The arm is the physical distance along the longitudinal axis from the established reference datum to the center of gravity of the specific item.

Q2: If an aircraft has a total mass of 2,000 kg and a total moment of 6,000 kg.m about the datum, where is the CG?

- [A]: 4.0 m aft of datum
- [B]: 3.0 m aft of datum
- [C]: 0.33 m aft of datum
- [D]: At the datum

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. $\text{CG} = \text{Total Moment} / \text{Total Mass} = 6,000 \text{ kg.m} / 2,000 \text{ kg} = 3.0 \text{ meters aft of the datum.}$

Q3: If a designer chooses the aircraft nose as the Datum, what sign will all arms of items loaded inside the aircraft have?

- [A]: Negative
- [B]: Positive (+), because all loaded items are located aft of the nose tip
- [C]: Zero
- [D]: Variable depending on flight speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When the datum is placed at the extreme front tip (nose), all positions in the cabin, wings, and empennage lie aft of the datum, meaning all arms and moments are positive (+), eliminating negative numbers.